Local Nonlinear Projection-Based Reduced-Order Models

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Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^{\star} = \arg\min_{\mathbf{q}} \; \frac{1}{2} \left\| \mathbf{r}(\mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q})) \right\|_2^2 .$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta\mathbf{q}$ is obtained from

$$\left(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi} \right) \Delta\mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}}, \qquad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \qquad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta\mathbf{q}.$$

Global Linear HPROM

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q}, \qquad \mathbf{V} \in \mathbb{R}^{N \times n},\ n = 151.$$

Hence

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \qquad \boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$\left(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}\right) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right), \qquad n = 16.$$

Thus

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}\left[\mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right)\right].$$

Global HPROM-GPR

Split basis and reduced coordinates:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \qquad \mathbf{q}_{\text{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\text{GPR}}(\mathbf{q}), \qquad n = 10,\ \bar{n} = 141.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}),$$

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\text{tot}}\mathbf{q}_{\text{tot}} = \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\,\mathcal{N}_{\text{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}}\frac{\partial \mathcal{N}_{\text{GPR}}}{\partial \mathbf{q}}.$$

Global HPROM-POD-AE

Latent reduced coordinate $\mathbf{z} \in \mathbb{R}^5$ with learned map:

$$\mathbf{q}_{\mathrm{tot}} = \mathcal{N}_{\mathrm{AE}}(\mathbf{z}), \qquad \mathbf{q}_{\mathrm{tot}} \in \mathbb{R}^{151}.$$

State reconstruction:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{z}),$$

$$\widetilde{\mathbf{u}}(\mathbf{z}) = \mathbf{V}_{\mathrm{tot}}\mathbf{q}_{\mathrm{tot}} = \mathbf{V}_{\mathrm{tot}}\mathcal{N}_{\mathrm{AE}}(\mathbf{z}), \qquad \mathbf{V}_{\mathrm{tot}} \in \mathbb{R}^{N\times 151}.$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{z}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{z}} = \mathbf{V}_{\mathrm{tot}}\frac{\partial \mathcal{N}_{\mathrm{AE}}}{\partial \mathbf{z}}.$$

Partition the training snapshots into clusters $k = 1, \dots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\mathrm{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k\mathbf{q},$$

$$\Phi_k^{\mathrm{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k\mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\mathrm{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k\mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\mathrm{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

In the next slides, we focus only on steps 1 and 2.

Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^{+} = \arg \min_{\ell=1,\dots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \qquad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold $\Phi_{k^-}^{\mathrm{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$:

$$k^{+} = \arg \min_{\ell=1,\dots,N_c} S_{k^-,\ell}^{\mathrm{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-,\ell}^{\mathrm{lin}}(\mathbf{q}) = 2\,\mathbf{g}_{k^-,\ell}^T\mathbf{q} + d_{k^-,\ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-,\ell} := \mathbf{V}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \qquad d_{k^-,\ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cluster Selection: Final Form (HQPROM and HPROM-GPR)

HQPROM selector

$$k^{+} = \arg\min_{\ell} S^{\mathrm{quad}}_{k-,\ell}(\mathbf{q}_{k-}),$$

$$S^{\mathrm{quad}}_{k-,\ell}(\mathbf{q}) = 2\,\mathbf{g}^{T}_{k-,\ell}\mathbf{q} + 2\,\mathbf{m}^{T}_{k-,\ell}(\mathbf{q}\otimes\mathbf{q}) + d_{k-,\ell},$$

$$\mathbf{m}^{T}_{k-,\ell}(\mathbf{q}\otimes\mathbf{q}) := (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^{T}\mathbf{H}_{k-}(\mathbf{q}\otimes\mathbf{q}).$$

HPROM-GPR selector

$$k^{+} = \arg\min_{\ell} S^{\mathrm{gpr}}_{k-,\ell}(\mathbf{q}_{k-}),$$

$$S^{\mathrm{gpr}}_{k-,\ell}(\mathbf{q}) = 2\,\mathbf{g}^{T}_{k-,\ell}\mathbf{q} + 2\,\bar{\mathbf{g}}^{T}_{k-,\ell}\mathcal{N}_{\mathrm{GPR},k-}(\mathbf{q}) + d_{k-,\ell}.$$

$$\bar{\mathbf{q}}_{k-} := \mathcal{N}_{\mathrm{GPR},k-}(\mathbf{q}_{k-}), \qquad \bar{\mathbf{g}}_{k-,\ell} := \bar{\mathbf{V}}^{T}_{k-}(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}).$$

Cluster Change: Linear and Nonlinear Transfers

$$k^+ = k^- \Rightarrow \text{no transfer}, \qquad k^+ \neq k^- \Rightarrow \text{solve transfer problem}.$$

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

Linear local target

$$\mathbf{q}_{k^+}^{\mathrm{lin}} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - \left(\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q} \right) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\mathrm{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

Nonlinear local targets (HQPROM / HPROM-GPR)

$$\mathbf{q}_{k^+}^{\mathrm{nl}} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\mathrm{nl}}(\mathbf{q}) \right\|_2^2.$$

Cluster Change: Nonlinear Least Squares

For nonlinear local manifolds, define

$$\mathbf{r}_{\mathrm{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \qquad \mathbf{J}_\Phi(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

Gauss–Newton at iterate $\mathbf{q}^{(j)}$:

$$\left(\mathbf{J}_\Phi^T \mathbf{J}_\Phi\right) \Delta\mathbf{q} = -\mathbf{J}_\Phi^T \mathbf{r}_{\mathrm{tr}}(\mathbf{q}^{(j)}), \qquad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta\mathbf{q}.$$

Online Performance (250×250): $\boldsymbol{\mu} = (4.56, 0.019)$

Computational model	n (N for HDM)	$\bar{n}$	N_e	$\mathbb{RE}_{2,\mathbf{u}}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	62 500	–	12.512	–
HPROM	151	–	5 722	0.450	1.262	9.913
HQPROM	19	–	1 588	5.085	0.362	34.541
HPROM-GPR	10	141	901	2.103	0.233	53.733
Local HPROM ($N_c = 10$)	24–47	–	2 951	0.228	0.380	32.934
Local HQPROM ($N_c = 10$)	6–9	–	640	2.269	0.258	48.550
Local HPROM-GPR ($N_c = 10$)	5	24–47	451	1.008	0.262	47.673
HPROM-POD-AE	5	151	451	1.422	0.208	60.291

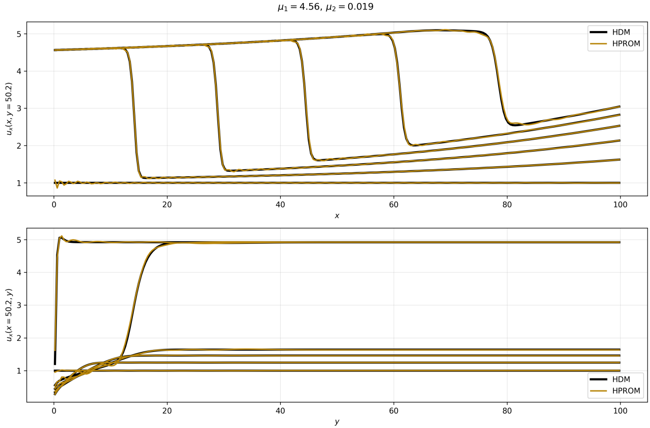
Speedup is computed with respect to the HDM runtime, $t_{\mathrm{HDM}} = 750.702$ s.

HPROM-POD-AE: $n = 5$ (latent) and $\bar{n} = n_{\mathrm{tot}} = 151$ (decoded coordinates).

N_e denotes the number of sampled ECSW elements ($N_e = 62{,}500$ for HDM/full mesh).

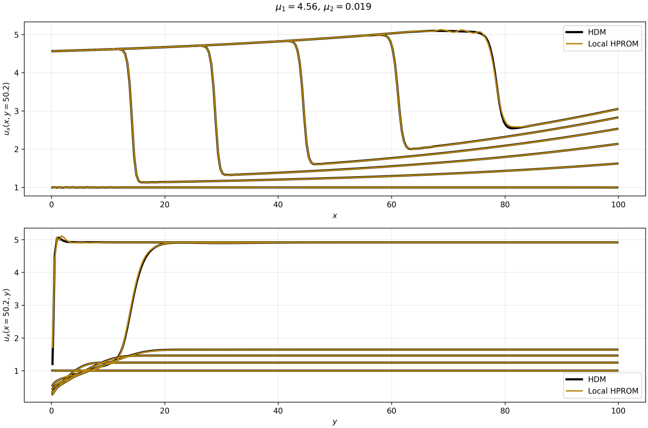
All local models use $N_c = 10$ clusters.

HPROM Solution (250×250): $\boldsymbol{\mu} = (4.56, 0.019)$



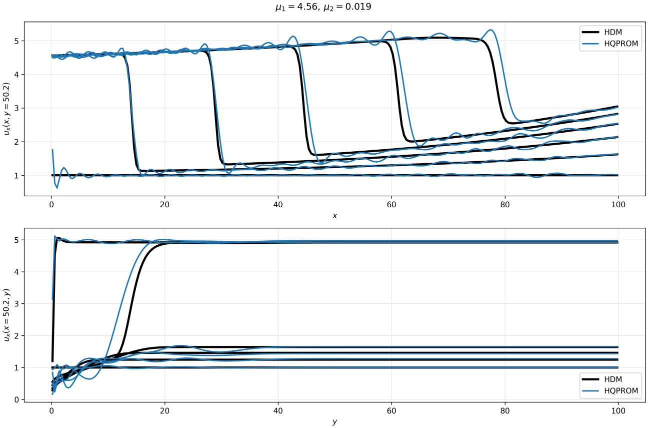
$\mathbb{RE}_{2,\mathbf{u}} = 0.450\%$ $\quad t = 75.731$ s (1.262 min) $\quad$ speedup $= 9.913\times$ $\quad N_e = 5{,}722$

Local HPROM Solution (250 × 250): $\boldsymbol{\mu} = (4.56, 0.019)$



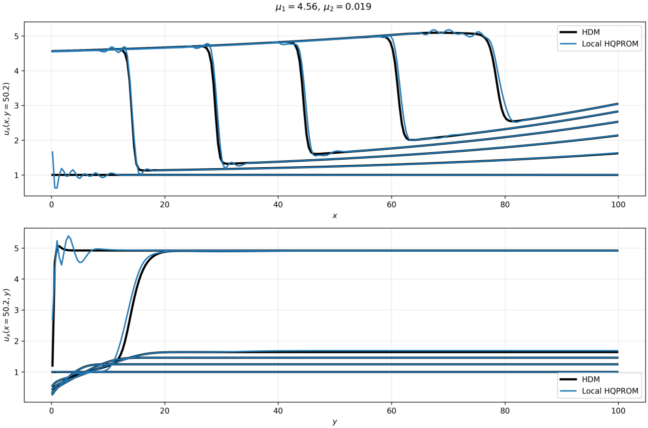
$\mathbb{RE}_{2,\mathbf{u}} = 0.228\%$ $\quad t = 22.794$ s (0.380 min) $\quad$ speedup $= 32.934\times$ $\quad N_e = 2{,}951$

HQPROM Solution (250 $\times$ 250): $\boldsymbol{\mu} = (4.56, 0.019)$



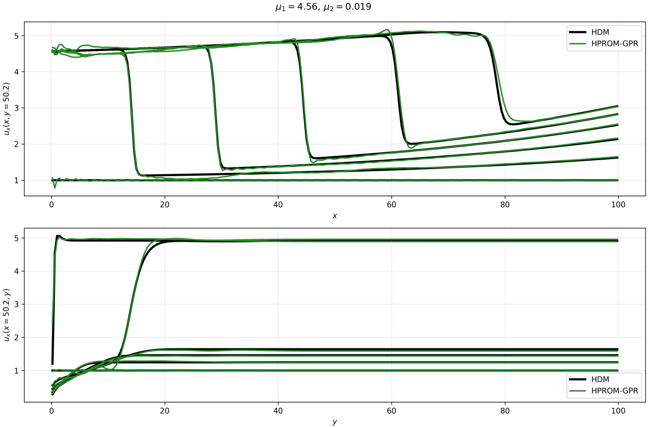
$\mathbb{RE}_{2,\mathbf{u}} = 5.085\%$ $t = 21.734$ s (0.362 min) speedup $= 34.541\times$ $N_e = 1{,}588$

Local HQPROM Solution (250 × 250): $\boldsymbol{\mu} = (4.56, 0.019)$



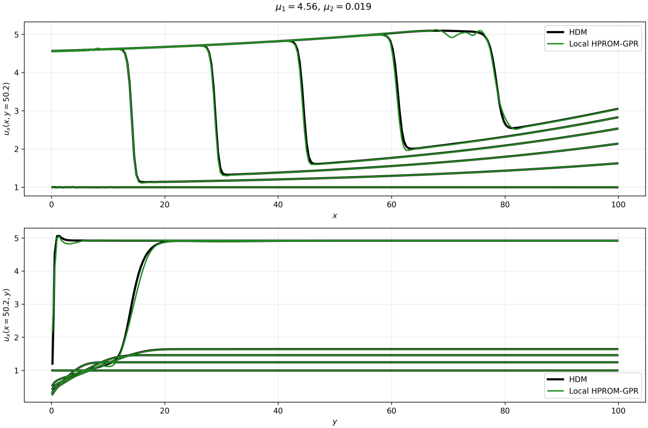
$\mathbb{RE}_{2,\mathbf{u}} = 2.269\%$ $\quad t = 15.463$ s (0.258 min) $\quad$ speedup $= 48.550\times$ $\quad N_e = 640$

HPROM-GPR Solution (250×250): $\boldsymbol{\mu} = (4.56, 0.019)$



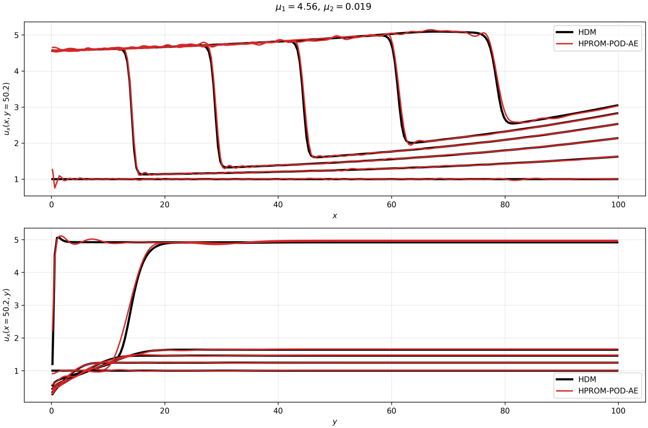
$\mathbb{RE}_{2,\mathbf{u}} = 2.103\%$ $t = 13.971$ s (0.233 min) speedup $= 53.733\times$ $N_e = 901$

Local HPROM-GPR Solution (250×250): $\boldsymbol{\mu} = (4.56, 0.019)$



$\mathbb{RE}_{2,\mathbf{u}} = 1.008\%$ $\quad t = 15.747$ s (0.262 min) $\quad$ speedup $= 47.673\times$ $\quad N_e = 451$

HPROM-POD-AE Solution (250×250): $\boldsymbol{\mu} = (4.56, 0.019)$



$\mathbb{RE}_{2,\mathbf{u}} = 1.422\%$ $\quad t = 12.451$ s (0.208 min) $\quad$ speedup $= 60.291\times$ $\quad N_e = 451$

Aero-F Implementation: Cylinder Case (Non-parametric)

Benchmark: 2D unsteady laminar viscous flow past a cylinder ($Re = 100$), non-parametric setup.

We report only HPROM-family results from the Aero-F workflow, using HDM as reference.

Reported metric

- Drag error: relative ℓ_2 error (%) of drag L_x over $t \in [0, 150]$.

Aero-F ROM tutorial repository (non-parametric cylinder case):

`https://github.com/SADPR/aero-f_rom_turorial`

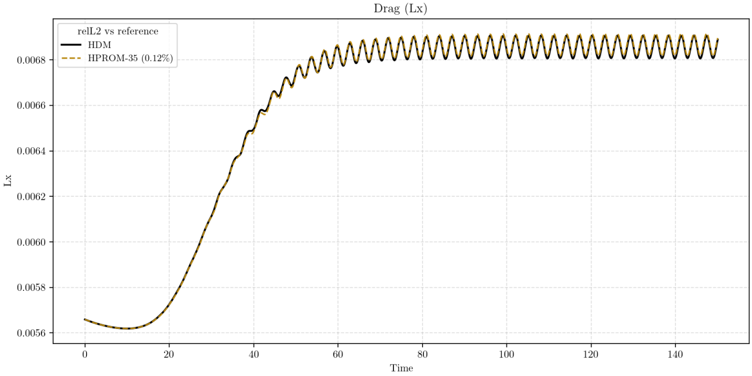
Runtimes were measured on a personal computer (not Sherlock), so wall-clock values are reference-only and may include parallel/background-load noise.

Aero-F Cylinder: HPROM Results Summary

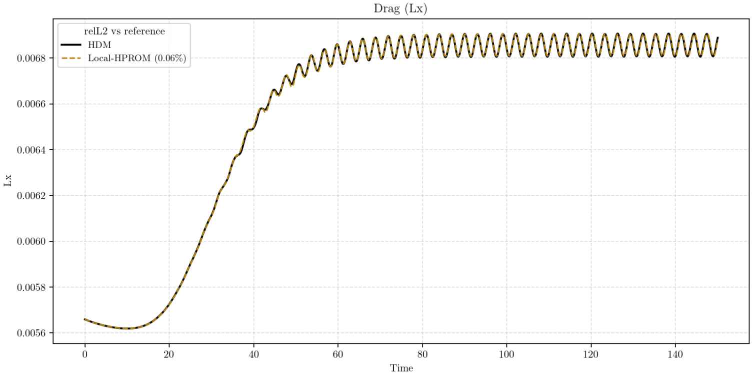
Computational model	n (N for HDM)	$\bar{n}$	N_e	$\mathbb{RE}_{2,L_x}$ (%)	Wall-clock time (min)	Speedup
HDM	490 700	–	98 140	–	218.622	–
HPROM	35	–	336	0.117	0.315	694.040
HQPROM	10	–	165	0.356	0.148	1 472.204
HPROM-ANN	5	30	93	0.159	0.092	2 367.751
HPROM-RBF	5	30	87	0.167	0.064	3 407.101
HPROM-GPR	5	30	81	0.134	0.058	3 769.351
Local-HPROM ($N_c = 3$)	14–35	–	253	0.062	0.179	1 223.632
Local-HQPROM ($N_c = 3$)	7–10	–	157	0.730	0.106	2 068.981
Local-HPROM-ANN ($N_c = 3$)	3	11–32	56	0.382	0.066	3 329.274
Local-HPROM-RBF ($N_c = 3$)	3	11–32	73	0.145	0.036	6 044.857
Local-HPROM-GPR ($N_c = 3$)	3	11–32	53	0.371	0.029	7 670.959

Drag error: relative ℓ_2 (%) of L_x on $t \in [0, 150]$. Speedup uses $t_{\mathrm{HDM}} = 13{,}117.340$ s. Global ANN/RBF/GPR use $n = 5$, $\bar{n} = 30$. Local ANN/RBF/GPR use $n = 3$, $\bar{n} \in [11, 32]$.

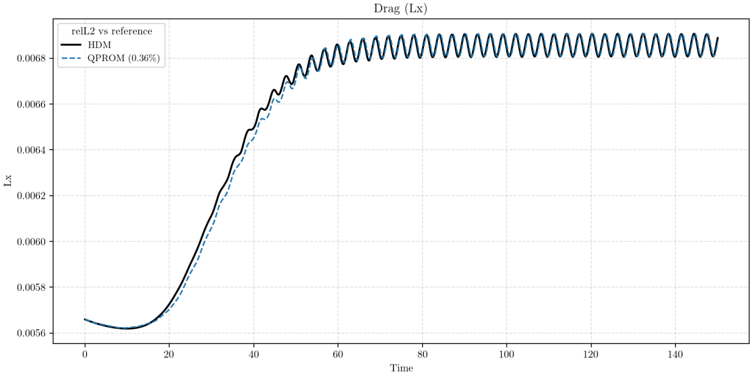
Aero-F Cylinder: HPROM vs HDM (Drag L_x)



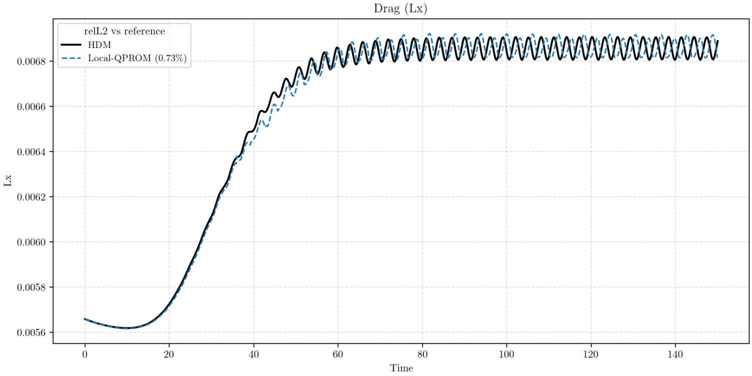
Aero-F Cylinder: Local-HPROM ($N_c = 3$) vs HDM (Drag L_x)



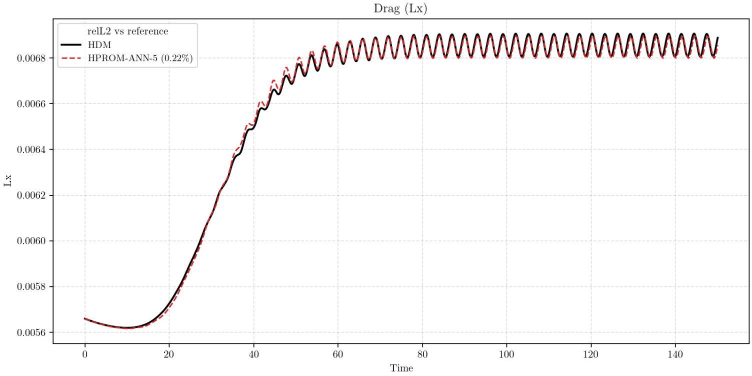
Aero-F Cylinder: HQPROM vs HDM (Drag L_x)



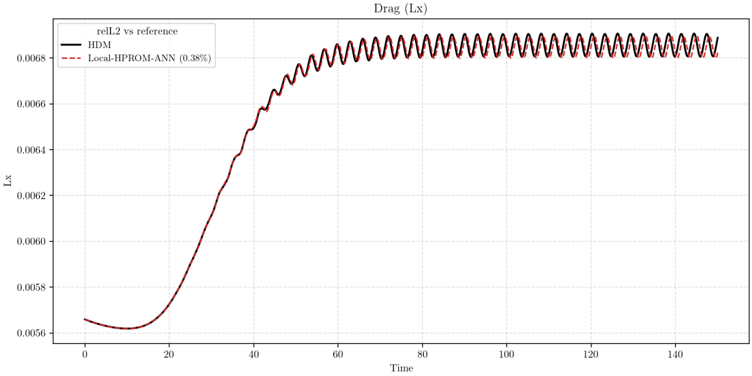
Aero-F Cylinder: Local-HQPROM ($N_c = 3$) vs HDM (Drag L_x)



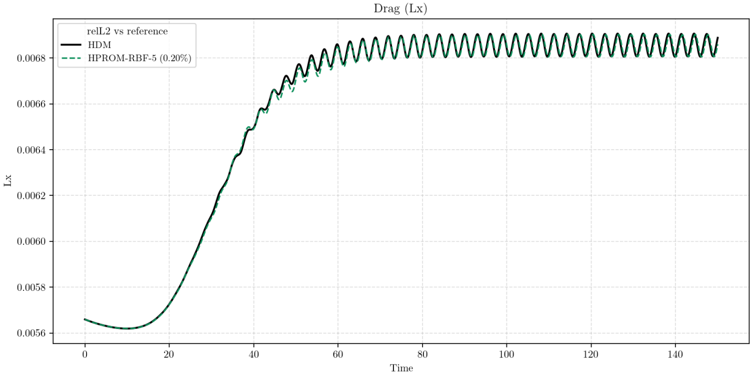
Aero-F Cylinder: HPROM-ANN vs HDM (Drag L_x)



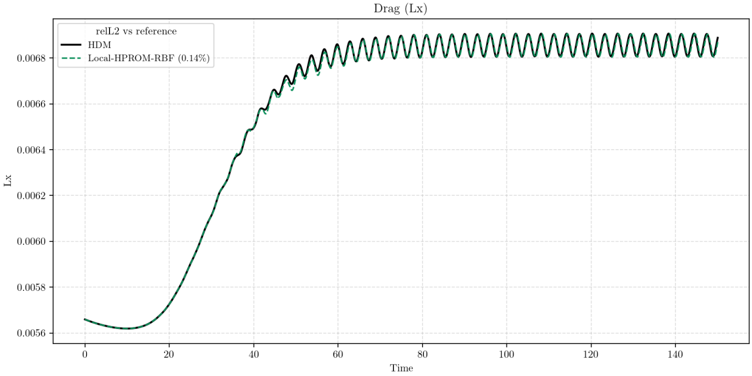
Aero-F Cylinder: Local-HPROM-ANN ($N_c = 3$) vs HDM (Drag L_x)



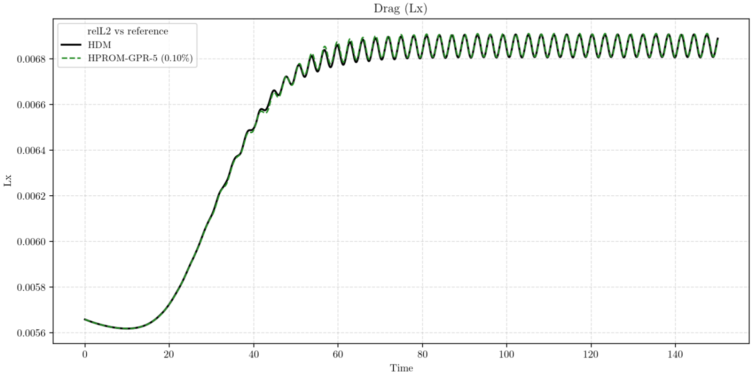
Aero-F Cylinder: HPROM-RBF vs HDM (Drag L_x)



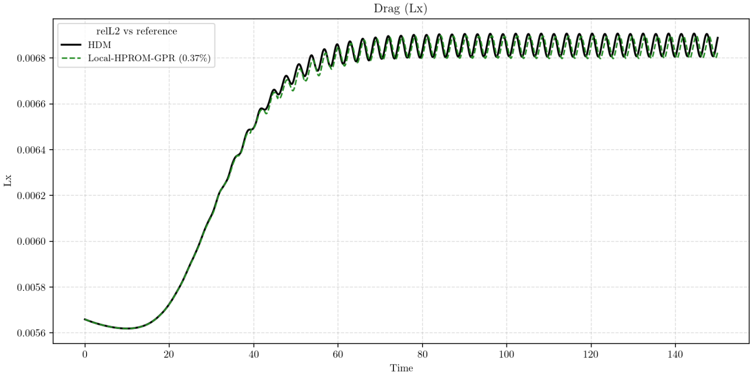
Aero-F Cylinder: Local-HPROM-RBF ($N_c = 3$) vs HDM (Drag L_x)



Aero-F Cylinder: HPROM-GPR vs HDM (Drag L_x)



Aero-F Cylinder: Local-HPROM-GPR ($N_c = 3$) vs HDM (Drag L_x)



Appendix: Linear Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-}\mathbf{q}_{k-}, \qquad k^+ = \arg\min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{q}_{k-}\|_2^2 + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k-}\mathbf{q}_{k-} + \|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Drop the ℓ-independent term $\|\mathbf{q}_{k-}\|_2^2$ and precompute

$$\mathbf{g}_{k-,\ell} := \mathbf{V}_{k-}^T(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}), \qquad d_{k-,\ell} := \|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2,$$

which yields

$$k^+ = \arg\min_{\ell} \left(2\,\mathbf{g}_{k-,\ell}^T\mathbf{q}_{k-} + d_{k-,\ell}\right).$$

Appendix: HQPROM Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-}\mathbf{q}_{k-} + \mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}).$$

$$k^+ = \arg\min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \arg\min_{\ell} \Big[\|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2$$

$$+2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k-}\mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k-}(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}) \Big].$$

Define

$$\mathbf{m}_{k-,\ell}^T(\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k-}(\mathbf{q} \otimes \mathbf{q}),$$

then

$$k^+ = \arg\min_{\ell} \left(2\,\mathbf{g}_{k-,\ell}^T \mathbf{q}_{k-} + 2\,\mathbf{m}_{k-,\ell}^T(\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}) + d_{k-,\ell} \right).$$

Appendix: HPROM-GPR Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}, \qquad \bar{\mathbf{q}}_{k^-} = \mathcal{N}_{\mathrm{GPR},k^-}(\mathbf{q}_{k^-}).$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k^-}\mathbf{q}_{k^-}$$

$$+2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-} + (\ell\text{-independent terms}).$$

Hence, with

$$\bar{\mathbf{g}}_{k^-,\ell} := \bar{\mathbf{V}}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}),$$

$$k^+ = \arg\min_{\ell}\left(2\,\mathbf{g}_{k^-,\ell}^T\mathbf{q}_{k^-} + 2\,\bar{\mathbf{g}}_{k^-,\ell}^T\bar{\mathbf{q}}_{k^-} + d_{k^-,\ell}\right).$$